

University Center of Illizi
Department of Computer Science

Homework No. 1, Analysis 1
Level: First year Bachelor's Degree

Exercise 1 Let $n \in \mathbb{N}^*$. Prove that

$$\forall a, b \in \mathbb{R} : \quad a^n - b^n = (a - b)(a^{n-1} + a^{n-2}b + a^{n-3}b^2 + \dots + b^{n-1}).$$

Exercise 2 We denote by $E(x)$ the greatest integer function of x , ($x \in \mathbb{R}$).

Prove that:

1. $\exists x, y \in \mathbb{R} : E(x + y) \neq E(x) + E(y)$,
2. $\exists x, y \in \mathbb{R} : E(x \cdot y) \neq E(x) \cdot E(y)$,
2. $\forall x \in \mathbb{R}$

$$E(x) + E(-x) = \begin{cases} 0, & \text{if } x \in \mathbb{Z}, \\ -1, & \text{if else.} \end{cases}$$

Exercise 3 Let $x_1, x_1, \dots, x_n \in \mathbb{R}$. Prove that

$$\left| \sum_{k=1}^n x_k \right| \leq \sum_{k=1}^n |x_k|.$$

Exercise 4 In $\mathbb{C}$ solve the following equations:

- 1). $|z + 3i| = |3z|$
- 2). $z^2 + 18z + 1681 = 0$.
- 3). $z^3 - z^2 + z - 1 = 0$.

Exercise 5 Let $\theta \in \mathbb{R}$ and $n \in \mathbb{N}^*$. Prove that

$$\cos^n \theta = \frac{1}{2^n} \sum_{k=0}^n C_n^k \cos(n - 2k)\theta,$$

$$\text{where } C_n^k = \binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Good luck
Dr. Zaid Laadjal